TECHNICAL ARCHITECTURE · PHASE 1

How we build it
AI Video Editor

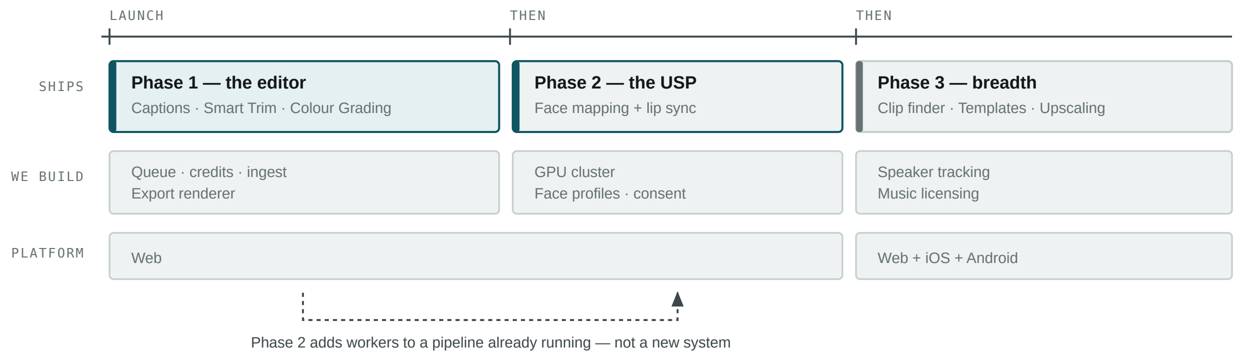
A web video editor with a real timeline, where the AI tools sit in the toolbar next to the ordinary ones. This page shows **how the system is put together** and the three decisions we still need from you.

EDITOR CONFIRMED | PHASED RELEASE | WEB FIRST | 12 AUGUST 2026

THE PLAN

Three phases, each building on the last

Phase 1 ships a working editor with the three AI tools that cost least and help most. The differentiator — face mapping — arrives in phase 2, onto machinery that is already running in production rather than being built alongside it.

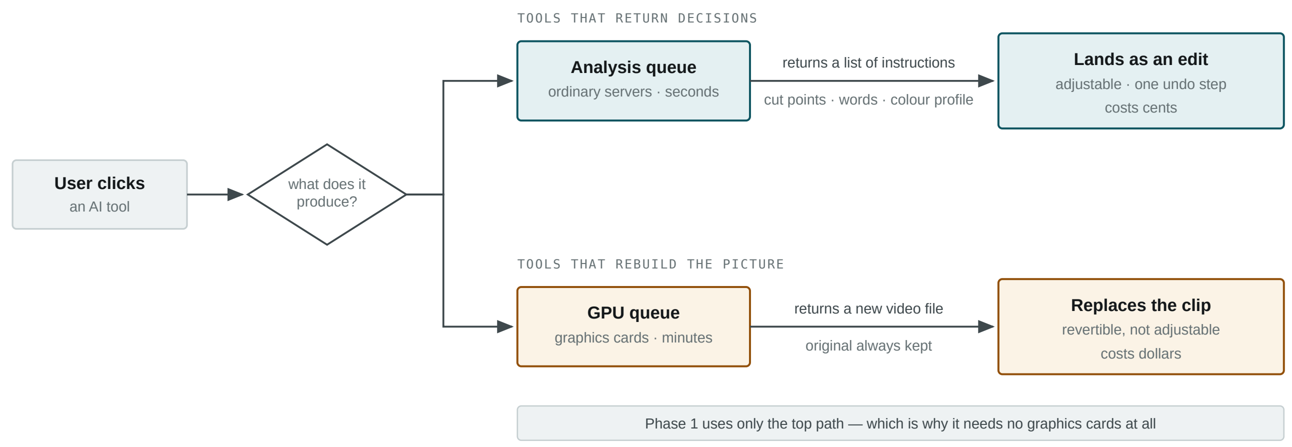


The roadmap read as a timeline. Everything in the middle row is built once, in phase 1, and reused by every phase after it. That is the whole argument for this ordering: phase 2 is the hardest AI feature, and it should not also be the first time we run a job queue in production.

THE KEY IDEA

AI tools come in two kinds, and it decides almost everything

Some tools can tell us **what to do** — where to cut, what was said, which colour profile fits. Others have to rebuild the picture itself. Those two things cost different amounts, take different hardware, and leave the user with different amounts of control. Sorting every tool into the right one is the single most consequential decision in the design.



Same button to the user, completely different machine underneath. Captions, smart trimming and colour grading all take the top path: they analyse and report back, cheaply, and the user can still change anything afterwards. Face mapping has no choice but the bottom path — there is no way to describe a replaced face as an instruction.

TOP PATH — PHASE 1

Cheap, fast, still editable

The server sends back a list. The editor turns it into ordinary cuts and captions that the user can drag, retype or undo.

TOOLS	Captions, Smart Trim, Colour Grading
TAKES	Seconds
NEEDS	Ordinary servers

BOTTOM PATH — PHASE 2

Slow, expensive, worth it

The server rebuilds the footage frame by frame and hands back a new file. The original is kept, so it can always be undone.

TOOLS	Face mapping, lip sync, noise removal
TAKES	Minutes
NEEDS	Graphics cards

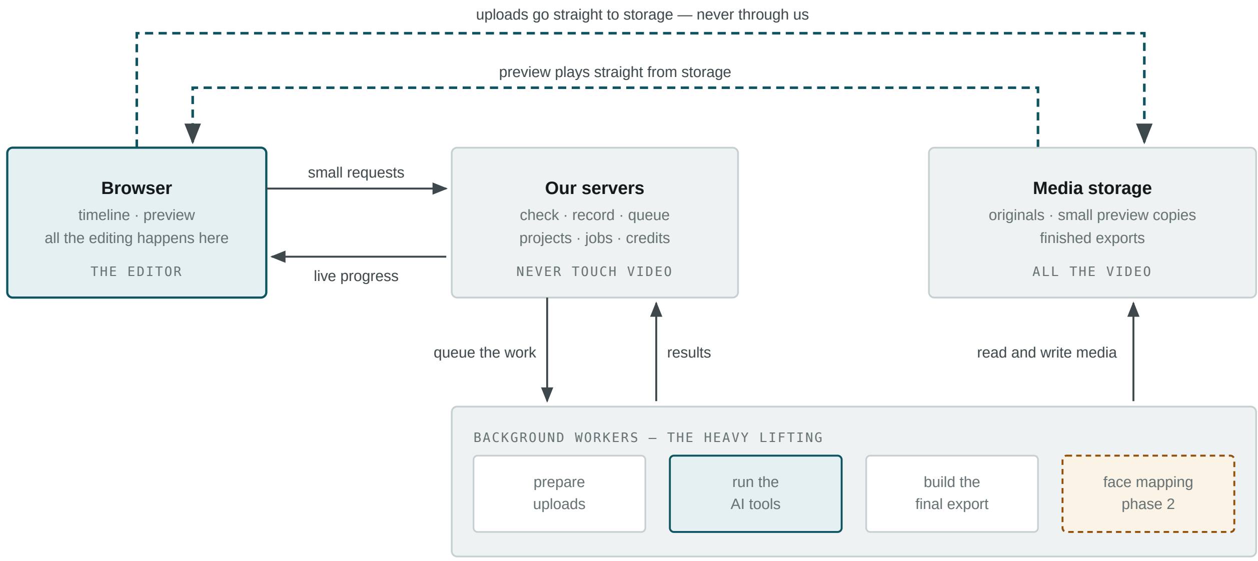
WHY THIS SAVES REAL MONEY

Graphics cards are rented by the hour whether they are working or not. By putting every phase 1 tool on the top path, we launch without renting any — and the cluster arrives in phase 2 alongside the one feature that genuinely needs it.

THE SYSTEM

What talks to what

The part worth noticing: video files never pass through our application. The browser uploads straight to storage and plays straight from the delivery network. Only small control messages — "make me a project", "run captions on this clip" — touch our servers.

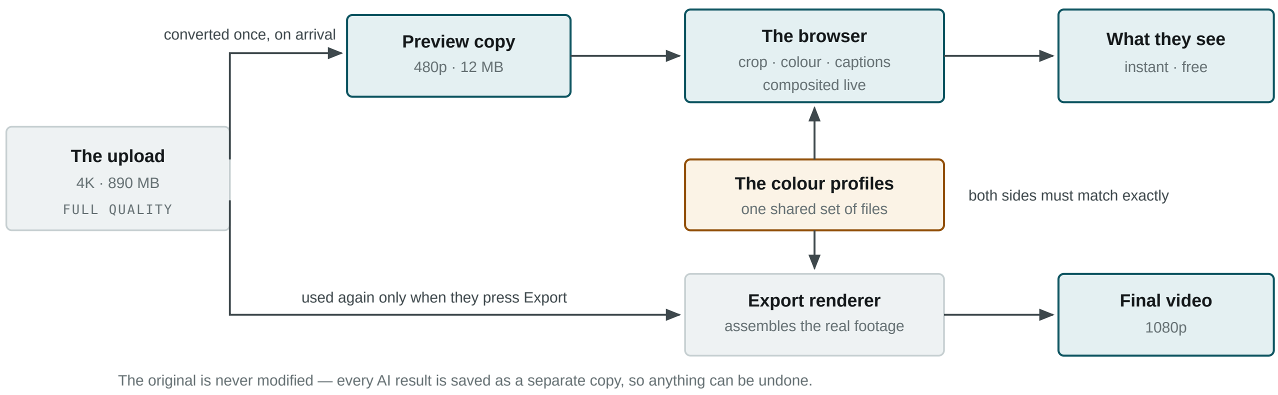


Dashed teal lines are video; solid lines are messages. Keeping video off our servers is what stops bandwidth costs scaling with usage — and it is why uploading a 900 MB file is as reliable as any other file transfer, rather than depending on our application staying up for ten minutes.

WHY IT FEELS INSTANT

Two copies of every video, doing two different jobs

When someone uploads footage, we immediately make a small, light copy of it. That copy is what they scrub, cut and preview — which is why editing responds instantly and costs us nothing. The full-quality original is touched exactly once more: at export.



The amber box is the part that has to be watched. The browser and the export renderer are two different pieces of software applying the same colour treatment. If they ever drift apart, someone edits a video that looks one way and downloads one that looks another — so they share the same profile files and are tested against each other.

WHAT WE NEED FROM YOU

Three decisions, and only the first one blocks us

Everything above can be built as described. These three are choices only you can make, and we would rather ask now than guess and rework later.

A

Confirm the three phase 1 tools: captions, smart trimming, colour grading

You said "2–3 core AI tools" — these are our proposal. All three are on the cheap path, all three answer a complaint creators actually have, and choosing them means launching without renting graphics cards.

Blocks: which worker gets written first, and whether phase 1 needs GPU hardware at all. Swapping one for another changes what we build first, not how anything is built.

B

Choose a payment provider and what each tool costs in credits

There is no payment system anywhere in the original architecture. The credit machinery is built either way — but nobody can buy anything until this exists.

Blocks: taking money. Does not block building or testing, which run on manually granted credits.

C

Smart trimming: tighten a recording, or cut it down to its best parts?

Removing silences and stumbles saves roughly 10–25% of a recording. The spec promises ten minutes down to two, which means choosing what is worth keeping — a substantially harder tool.

Blocks: whether smart trimming is in phase 1 at all. If it is the harder version, we would swap it out and revisit it later.

ONE THING TO KEEP IN VIEW

Phase 1 ships without face mapping, so it cannot be marketed as the face-swap product — its story is "the editor that does the boring work for you". That is the right build order, but if launch is being sold on the differentiator, it is a conversation worth having now rather than in three months.

THE FULL SET

Where the detail lives

This page is the overview. Everything below is written and in the repository, ready for the development team.

01

Product Vision

What the product does, every decision made and every one still open

02

Phase 1 Scope

The cold list of what ships now, and what explicitly does not

03

Backend Architecture

Data model, job pipeline, media handling, rendering, infrastructure

04

Frontend Architecture

Timeline state, the playback engine, undo, tool integration

05

API Contract

The interface both teams build against, in full

—

Diagrams

System overview, data model, job lifecycle

Built on the project lead's decisions of 12 August 2026. Phase 1 scope is firm; the AI tool selection is a proposal awaiting decision A.
Full documentation set in `docs/`.